Package mcst: Sample Scenario Illustrations

Practical Examples of Simulation, Stress Testing, and Results Extraction

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1 Installation of package

The latest development version of the package can be installed using `remotes::install_github("calebong/mcst" dependencies = TRUE)`

```r
require(remotes)
remotes::install_github("calebong/mcst", dependencies = TRUE)
```

```
## These packages have more recent versions available.
## It is recommended to update all of them.
## Which would you like to update?
##
##  1: All
##  2: CRAN packages only
##  3: None
##  4: rlang        (1.1.7      -> 1.2.0   ) [CRAN]
##  5: rappdirs     (0.3.3      -> 0.3.4   ) [CRAN]
##  6: highr        (0.11       -> 0.12    ) [CRAN]
##  7: digest       (0.6.37     -> 0.6.39  ) [CRAN]
##  8: sass         (0.4.9      -> 0.4.10  ) [CRAN]
##  9: mime         (0.12       -> 0.13    ) [CRAN]
## 10: base64enc    (0.1-3      -> 0.1-6   ) [CRAN]
## 11: yaml         (2.3.10     -> 2.3.12  ) [CRAN]
## 12: xfun         (0.50       -> 0.57    ) [CRAN]
## 13: tinytex      (0.54       -> 0.59    ) [CRAN]
## 14: knitr        (1.49       -> 1.51    ) [CRAN]
## 15: htmltools    (0.5.8.1    -> 0.5.9   ) [CRAN]
## 16: bslib        (0.9.0      -> 0.10.0  ) [CRAN]
## 17: stringi      (1.8.4      -> 1.8.7   ) [CRAN]
## 18: textshaping  (0.3.7      -> 1.0.5   ) [CRAN]
## 19: systemfonts  (1.0.5      -> 1.3.2   ) [CRAN]
## 20: vctrs        (0.6.5      -> 0.7.2   ) [CRAN]
## 21: openssl      (2.3.2      -> 2.3.5   ) [CRAN]
## 22: magrittr     (2.0.3      -> 2.0.5   ) [CRAN]
## 23: curl         (5.2.0      -> 7.0.0   ) [CRAN]
## 24: fansi        (1.0.6      -> 1.0.7   ) [CRAN]
## 25: diffobj      (0.3.5      -> 0.3.6   ) [CRAN]
## 26: utf8         (1.2.4      -> 1.2.6   ) [CRAN]
## 27: tibble       (3.2.1      -> 3.3.1   ) [CRAN]
## 28: pillar       (1.10.1     -> 1.11.1  ) [CRAN]
## 29: generics     (0.1.3      -> 0.1.4   ) [CRAN]
## 30: nloptr       (2.0.3      -> 2.2.1   ) [CRAN]
## 31: Rcpp         (1.0.14     -> 1.1.1   ) [CRAN]
## 32: Distribut... (0.6-1      -> 0.6-2   ) [CRAN]
## 33: zoo          (1.8-12     -> 1.8-15  ) [CRAN]
## 34: mvtnorm      (1.2-4      -> 1.3-6   ) [CRAN]
## 35: multicool    (1.0.0      -> 1.0.1   ) [CRAN]
## 36: mclust       (6.0.1      -> 6.1.2   ) [CRAN]
```

```
## 37: kernlab      (0.9-32     -> 0.9-33  ) [CRAN]
## 38: FNN          (1.1.3.2    -> 1.1.4.1 ) [CRAN]
## 39: parallelly   (1.42.0     -> 1.46.1  ) [CRAN]
## 40: listenv      (0.9.1      -> 0.10.1  ) [CRAN]
## 41: globals      (0.16.3     -> 0.19.1  ) [CRAN]
## 42: future       (1.34.0     -> 1.70.0  ) [CRAN]
## 43: RcppArmad... (0.12.6.6.1 -> 15.2.4-1) [CRAN]
## 44: future.apply (1.11.3     -> 1.20.2  ) [CRAN]
## 45: viridisLite  (0.4.2      -> 0.4.3   ) [CRAN]
## 46: isoband      (0.2.7      -> 0.3.0   ) [CRAN]
## 47: xml2         (1.3.6      -> 1.5.2   ) [CRAN]
## 48: rmarkdown    (2.29       -> 2.31    ) [CRAN]
## 49: ragg         (1.2.7      -> 1.5.2   ) [CRAN]
## 50: purrr        (1.0.4      -> 1.2.1   ) [CRAN]
## 51: httr2        (1.0.0      -> 1.2.2   ) [CRAN]
## 52: stringr      (1.5.1      -> 1.6.0   ) [CRAN]
## 53: dplyr        (1.1.4      -> 1.2.1   ) [CRAN]
## 54: fracdiff     (1.5-2      -> 1.5-3   ) [CRAN]
## 55: chron        (2.3-61     -> 2.3-62  ) [CRAN]
## 56: xts          (0.14.1     -> 0.14.2  ) [CRAN]
## 57: ks           (1.14.1     -> 1.15.1  ) [CRAN]
## 58: Rsolnp       (1.16       -> 2.0.1   ) [CRAN]
## 59: pspline      (1.0-19     -> 1.0-21  ) [CRAN]
## 60: pcaPP        (2.0-4      -> 2.0-5   ) [CRAN]
## 61: colorspace   (2.1-1      -> 2.1-2   ) [CRAN]
## 62: pkgdown      (2.0.7      -> 2.2.0   ) [CRAN]
## 63: tidyr        (1.3.1      -> 1.3.2   ) [CRAN]
## 64: rugarch      (1.5-1      -> 1.5-5   ) [CRAN]
## 65: copula       (1.1-3      -> 1.1-7   ) [CRAN]
## 66: patchwork    (1.3.0      -> 1.3.2   ) [CRAN]
##
## -- R CMD build ------------------------------------------------------------------------------
##      checking for file '/private/var/folders/p0/hl3w4hkn0yg0_dyzyhx62n440000gn/T/RtmpaZvq01/
##   -  preparing 'mcst':
##      checking DESCRIPTION meta-information ...  v  checking DESCRIPTION meta-information
##   -  checking for LF line-endings in source and make files and shell scripts
##   -  checking for empty or unneeded directories
##        NB: this package now depends on R (>= 3.5.0)
##        WARNING: Added dependency on R >= 3.5.0 because serialized objects in
##      serialize/load version 3 cannot be read in older versions of R.
##      File(s) containing such objects:
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/endo-data_d62566815bbc17a2e093a
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/endo-data_d62566815bbc17a2e093a
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/exo-data_5eb69e4e324e8cfb31a6f0
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/exo-data_5eb69e4e324e8cfb31a6f0
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s1-run_8df1b768431e7a7f7fc5b546
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s1-run_8df1b768431e7a7f7fc5b546
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s2-run_3cd7ba8a00e03cf5211f3c23
```

```
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s2-run_3cd7ba8a00e03cf5211f3c23
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s3-run_a08fa42e37301f833d7dc9c4
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s3-run_a08fa42e37301f833d7dc9c4
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s4-run_5918272ff732265f85a98646
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s4-run_5918272ff732265f85a98646
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s5-run_489aeab4a62538d8d78703d
##        'mcst/portfolio_risk_simulation_scenarios_cache/latex/s5-run_489aeab4a62538d8d78703d
## -  building 'mcst_0.1.0.tar.gz'
##      Warning in utils::tar(filepath, pkgname, compression = compression, compression_level =
##      storing paths of more than 100 bytes is not portable:
##      'mcst/portfolio_risk_simulation_scenarios_cache/latex/endo-data_d62566815bbc17a2e093ad4
##
##
```

2 Data Preparation

2.1 Portfolio Assets

For illustration we use the following packages to download 10 stocks from Yahoo finance at a daily frequency. The returns are then aggregated to weekly returns. These form the **endogenous** assets - the holdings to be simulated.

```r
# simple function to convert class xts to dataframe
xts_to_dataframe <- function(xts_obj, column_names = NULL) {
  dates <- zoo::index(xts_obj)
  df    <- as.data.frame(xts_obj)
  df    <- cbind(date = as.Date(dates), df)
  if (!is.null(column_names)) names(df)[-1] <- column_names
  rownames(df) <- NULL
  df
}
```

```r
# load in the libraries to get sample stocks data
library(tidyverse)
library(quantmod)
library(lubridate)

tickers <- c("NVDA","TSM","JPM","AXP","LMT","BA","UAL","XOM","NGG","PG")

out <- map(tickers, function(.x) {
  getSymbols(.x, from = "2010-01-01", to = Sys.Date(),
             src = "yahoo", auto.assign = FALSE) %>%
    xts_to_dataframe() %>%
    dplyr::select(1, 7) %>% #quickly select date and adjusted prices
    rename(Price = 2)
})
names(out) <- tickers

out_wide <- out %>%
```

```r
  enframe() %>%
  unnest(cols = c(value)) %>%
  pivot_wider(names_from = "name", values_from = "Price")

weekly_returns <- out_wide %>%
  pivot_longer(cols = -date, names_to = "sector", values_to = "price") %>%
  mutate(week = floor_date(date, unit = "week")) %>%
  group_by(sector, week) %>%
  summarise(price_week_end = last(price), .groups = "drop") %>%
  group_by(sector) %>%
  arrange(week) %>%
  mutate(weekly_return = price_week_end / lag(price_week_end) - 1) %>%
  ungroup() %>%
  filter(!is.na(weekly_return)) %>%
  dplyr::select(week, sector, weekly_return) %>%
  pivot_wider(id_cols = week, names_from = sector, values_from = weekly_return)
```

2.2 Exogenous Instruments

SPY (S&P 500 ETF - quick proxy for S&P 500 Index) and Crude Oil futures (`CL=F`) are downloaded separately as **exogenous** instruments. These carry zero portfolio weight and are used only as shock transmission channels in the stress-testing step.

```r
exo    <- c("SPY", "CL=F")
naming <- c("SPY", "Crude")

out_exo <- map(exo, function(.x) {
  getSymbols(.x, from = "2010-01-01", to = Sys.Date(),
             src = "yahoo", auto.assign = FALSE) %>%
    xts_to_dataframe() %>%
    dplyr::select(1, 7) %>% # quickly select date and adjusted prices
    rename(Price = 2)
})
names(out_exo) <- naming

out_wide_exo <- out_exo %>%
  enframe() %>%
  unnest(cols = c(value)) %>%
  pivot_wider(names_from = "name", values_from = "Price")

weekly_returns_exo <- out_wide_exo %>%
  pivot_longer(cols = -date, names_to = "sector", values_to = "price") %>%
  mutate(week = floor_date(date, unit = "week")) %>%
  group_by(sector, week) %>%
  summarise(price_week_end = last(price), .groups = "drop") %>%
  group_by(sector) %>%
  arrange(week) %>%
  mutate(weekly_return = price_week_end / lag(price_week_end) - 1) %>%
```

```r
  ungroup() %>%
  filter(!is.na(weekly_return)) %>%
  dplyr::select(week, sector, weekly_return) %>%
  pivot_wider(id_cols = week, names_from = sector, values_from = weekly_return) %>%
  mutate(Crude = ifelse(Crude <= -1, -0.5, Crude))
```

2.3 Combine and Filter

To produce a data frame quickly for illustration, portfolio and exogenous returns are joined on the week key and filtered to start from 2016, giving a cleaner estimation sample.

```r
comb <- weekly_returns %>%
  full_join(weekly_returns_exo, by = "week") %>%
  mutate(across(where(is.numeric), ~replace_na(., 0))) %>%
  filter(week >= "2016-01-01")

returns_data <- comb %>% dplyr::select(-Crude, -SPY)
exo_data     <- comb %>% dplyr::select(week, Crude, SPY)

cat(sprintf("Portfolio : %d assets, %d weekly obs\n",
            ncol(returns_data) - 1, nrow(returns_data)))
```

```
## Portfolio : 10 assets, 536 weekly obs
```

```r
cat(sprintf("Exogenous : %d instruments\n", ncol(exo_data) - 1))
```

```
## Exogenous : 2 instruments
```

```r
# data structure of input data frames
head(returns_data)
```

```
## # A tibble: 6 x 11
##   week            AXP       BA      JPM      LMT      NGG    NVDA       PG     TSM     UAL
##   <date>        <dbl>    <dbl>    <dbl>    <dbl>    <dbl>   <dbl>    <dbl>   <dbl>   <dbl>
## 1 2016-01-03 -0.0811  -0.101   -0.102   -0.0195  -0.0168  -0.101  -0.0433  -0.0853 -0.0944 -
## 2 2016-01-10 -0.0113  -0.0335  -0.0319   0.0145  -0.0212  -0.0850 -0.0130   0.0139 -0.120
## 3 2016-01-17 -0.125   -0.00812 -0.00158 -0.0183   0.0227   0.0494  0.0407   0.0393  0.0230 -
## 4 2016-01-24 -0.0283  -0.0360   0.0448  -0.00491  0.0346   0.0295  0.0560   0.0192  0.0334
## 5 2016-01-31  0.00897  0.0202  -0.0294   0.00445 -0.00946 -0.0976 -0.00600  0.0179 -0.0294
## 6 2016-02-07 -0.0245  -0.105   -0.00450 -0.0276  -0.0204  -0.0265 -0.00259 -0.0224  0.0318
```

```r
head(exo_data)
```

```
## # A tibble: 6 x 3
##   week         Crude      SPY
##   <date>       <dbl>    <dbl>
## 1 2016-01-03 -0.105  -0.0586
## 2 2016-01-10 -0.113  -0.0214
## 3 2016-01-17  0       0
## 4 2016-01-24  0       0
```

```
## 5 2016-01-31 -0.0812 -0.0298
## 6 2016-02-07 -0.0469 -0.00702
```

3 Scenario 1: Base Case (Gaussian, Equal Weights)

The simplest possible call. No arguments are overridden: the function uses a multivariate Normal distribution, historical sample covariance, and equal weights across all ten assets. No stress is applied.

```
library(mcst)

result_1 <- portfolio_risk_simulation(
  historical_returns = returns_data
)
```

3.1 Fitted Marginal Distributions

Each panel shows the empirical return histogram with the fitted Normal density overlaid.

```
result_1$plots$portfolio_diagnostics$asset_distributions
```

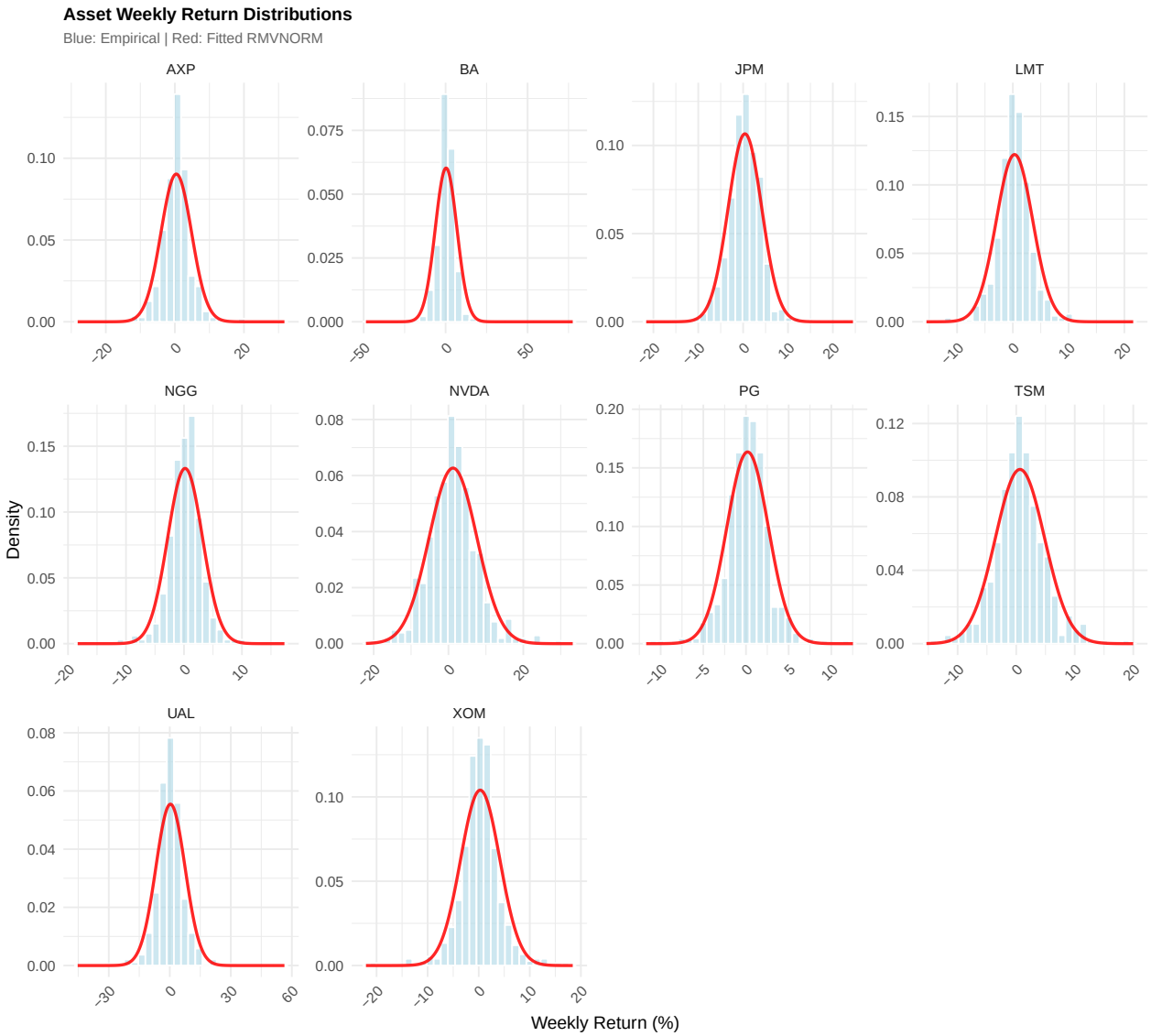

Figure 1: Scenario 1: Fitted Normal marginals vs. empirical returns per asset.

3.2 Simulated Portfolio Return Distribution

```
result_1$plots$portfolio_diagnostics$portfolio_distribution
```

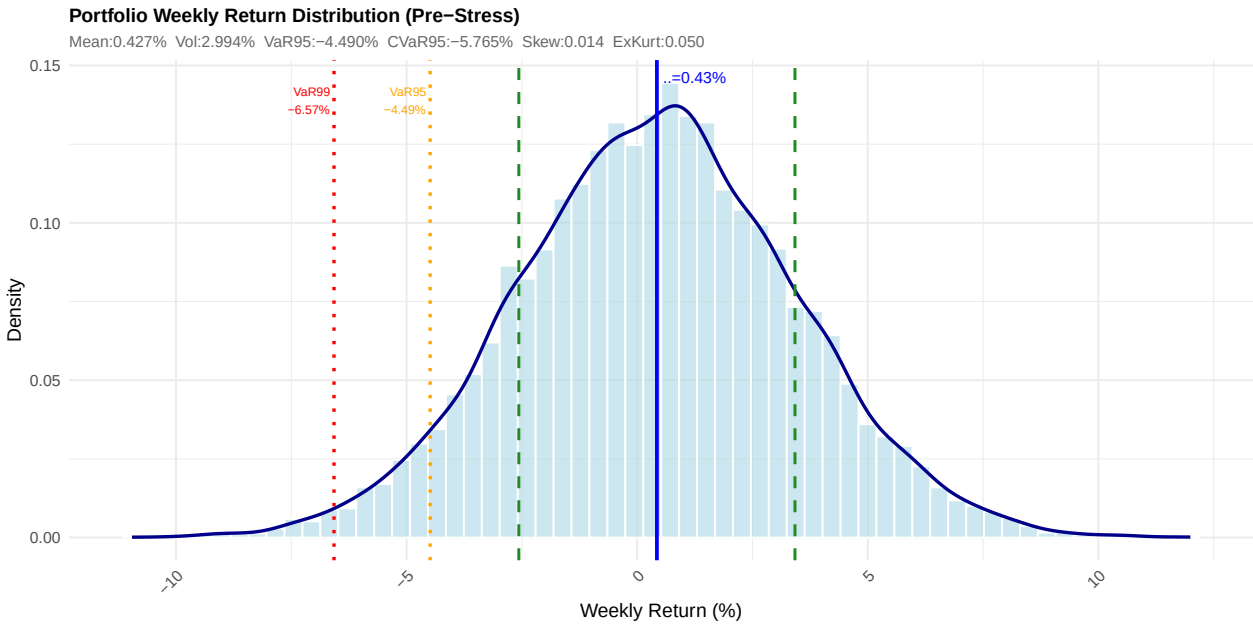

Figure 2: Scenario 1: Simulated portfolio return distribution (Gaussian, no stress).

3.3 Risk Metrics

Table 1: Scenario 1: Portfolio risk statistics (Gaussian base case).

Metric	Pre-Stress	Post-Stress
Mean (%)	0.4273	NA
Median (%)	0.4411	NA
Volatility (%)	2.9937	NA
Skewness	0.0141	NA
Excess Kurtosis	0.0496	NA
Sharpe (Annual)	1.0293	NA
Annual Return (%)	24.8225	NA
Annual Vol (%)	21.5876	NA
VaR 90% (%)	-3.3686	NA
VaR 95% (%)	-4.4897	NA
VaR 99% (%)	-6.5737	NA
CVaR 90% (%)	-4.8204	NA
CVaR 95% (%)	-5.7651	NA
CVaR 99% (%)	-7.5864	NA

3.4 Per-Asset Empirical Summary

Table 2: Scenario 1: Per-asset empirical return statistics.

Asset	Mean	SD	Skewness	Kurtosis	Min	Max
AXP	0.0040	0.0442	0.1545	10.2072	-0.2558	0.2883
BA	0.0030	0.0662	1.7068	33.6212	-0.4418	0.7051
JPM	0.0040	0.0374	-0.0035	7.4175	-0.1964	0.2226
LMT	0.0030	0.0327	0.2898	8.2515	-0.1409	0.1963
NGG	0.0019	0.0300	-0.5119	8.6672	-0.1667	0.1575

NVDA	0.0121	0.0637	0.3622	4.4982	-0.2005	0.3019
PG	0.0019	0.0244	-0.1781	5.7295	-0.1063	0.1140
TSM	0.0065	0.0420	0.2188	4.2185	-0.1393	0.1817
UAL	0.0034	0.0719	0.6876	12.2523	-0.4116	0.5125
XOM	0.0030	0.0384	-0.1108	6.6730	-0.2007	0.1674

4 Scenario 2: Fat Tails with t-Copula and EWMA Covariance

Changes from base case: Multivariate t returns (`rmvt`), t-copula to capture tail co-dependence, EWMA covariance ($\lambda = 0.97$) so recent observations receive higher weight, and per-asset degrees-of-freedom calibrated from the data (`calibration_method = "full"`).

```
result_2 <- portfolio_risk_simulation(
  historical_returns  = returns_data,
  sim_returns_dist    = "rmvt",
  cov_estimation      = "ewma",
  cov_lambda          = 0.97,
  calibrate_params    = TRUE,
  calibration_method  = "full",
  copula              = TRUE,
  copula_type         = "t",
  copula_df           = 4,
  n_sim_returns       = NULL,
  propagation         = TRUE,
  exogenous_returns   = NULL,
  exogenous_shock     = NULL,
  exogenous_volatility_shock = NULL,
  df                  = 4,
  seed                = 123
)
```

4.1 Tail Dependence and Copula Diagnostics

The left panel shows the pairwise lower tail dependence coefficient λ - the probability that two assets move to extreme values simultaneously. The right panel compares fitted copula correlations to the Kendall τ-implied starting values: points far from the diagonal indicate that MLE optimisation shifted that pair's correlation materially from the method-of-moments estimate.

```
result_2$plots$portfolio_diagnostics$tail_dependency
```

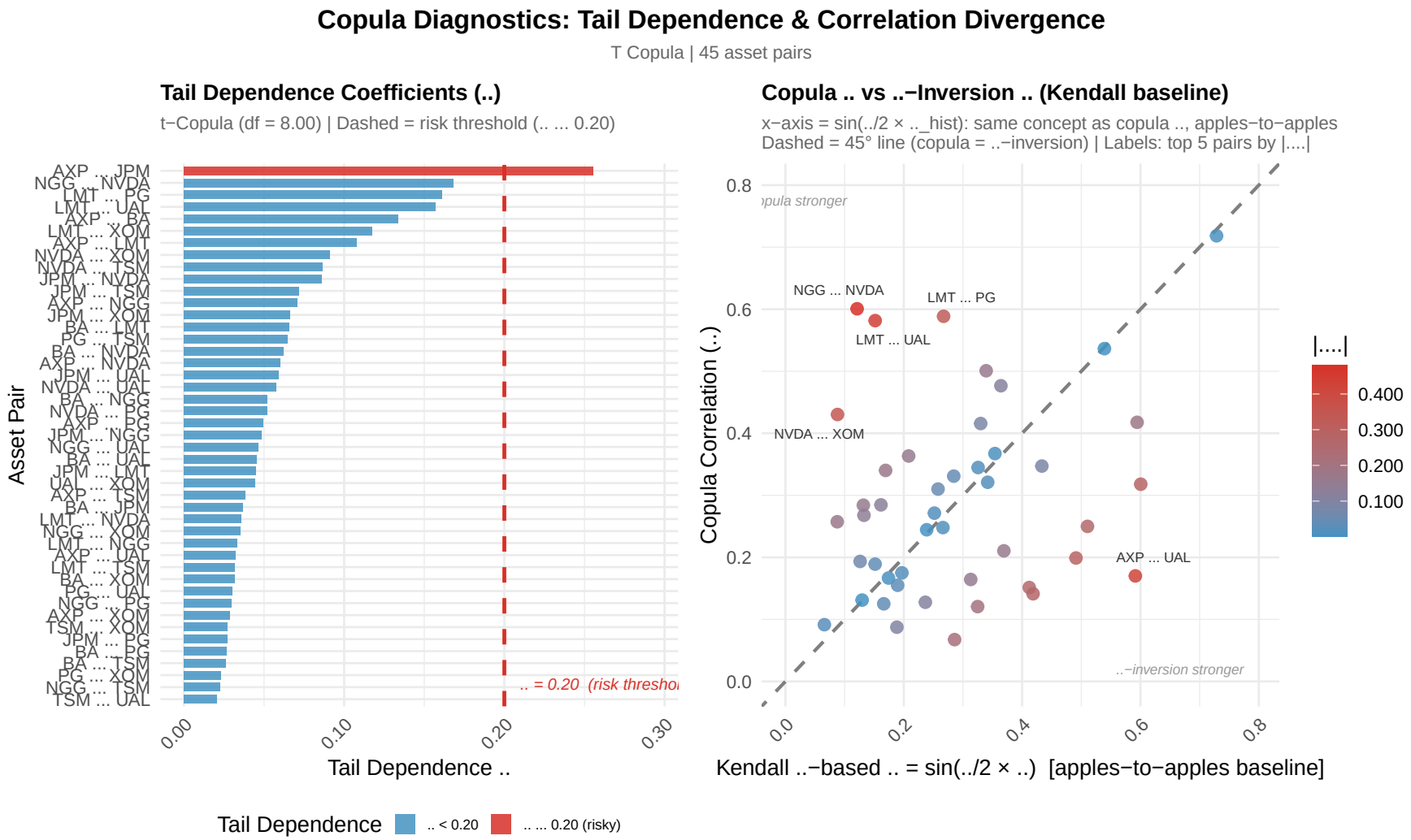

Figure 3: Scenario 2: Pairwise tail dependence and copula vs. Kendall baseline.

4.2 KS Uniformity Test

The KS test checks whether each asset's PIT residuals are Uniform[0,1] after the t-marginal is fitted. A rejection ($p < 0.05$) flags a misspecified marginal for that asset.

Table 3: Scenario 2: KS uniformity test on PIT residuals.

Asset	KS_stat	p_value	Flag
AXP	0.0673	0.0157	**REJECT**
BA	0.0988	0.0001	**REJECT**
JPM	0.0424	0.2896	**OK** |
LMT	0.0589	0.0486	**REJECT**
NGG	0.0519	0.1113	**OK** |
NVDA	0.0339	0.5694	**OK** |
PG	0.0489	0.1540	**OK** |
TSM	0.0364	0.4779	**OK** |
UAL	0.0708	0.0093	**REJECT**
XOM	0.0633	0.0271	**REJECT**

5 Scenario 3: Exogenous Volatility Shock - SPY Vol x2 (GJR-GARCH)

Shock: SPY volatility doubles ($k_{SPY} = 2$); Crude is unchanged ($k_{Crude} = 1$). No return shock is applied, so **portfolio means are unaffected** - only the covariance widens via the Law of Total Variance addon. GJR-GARCH estimates per-asset forward volatility, capturing volatility clustering and the leverage effect.

```
exogenous_volatility_shock <- list(SPY = 2, Crude = 1)
```

```
result_3 <- portfolio_risk_simulation(
  historical_returns          = returns_data,
  sim_returns_dist            = "rmvt",
  cov_estimation              = "garch",
  calibrate_params            = TRUE,
  calibration_method          = "full",
  copula                      = TRUE,
  copula_type                 = "t",
  copula_df                   = 4,
  n_sim_returns               = 10000,
  propagation                 = TRUE,
  exogenous_returns           = exo_data,
  exogenous_shock             = NULL,
  exogenous_volatility_shock  = exogenous_volatility_shock,
  df                          = 4,
  seed                        = 123
)
```

5.1 GJR-GARCH Conditional Volatility Paths

Each panel shows: salmon bars = raw returns; blue line = in-sample conditional $\hat{\sigma}_t$; red dashed = long-run unconditional $\bar{\sigma}$; green diamond = 1-step-ahead forecast. Assets where the forecast sits well above $\bar{\sigma}$ are in an elevated-volatility regime.

```r
result_3$plots$portfolio_diagnostics$time_series
```

Figure 4: Scenario 3: GJR-GARCH conditional volatility paths per asset.

5.2 Exogenous Propagation Channel

The beta $\beta_{pe} = \mathrm{Cov}(X_p, X_e)/\mathrm{Var}(X_e)$ quantifies the transmission of SPY volatility to each portfolio asset. With a pure vol shock (no return shock), these betas determine the size of the LTV covariance addon - not any mean shift.

```
result_3$stressed_plots$exogenous_propagation
```

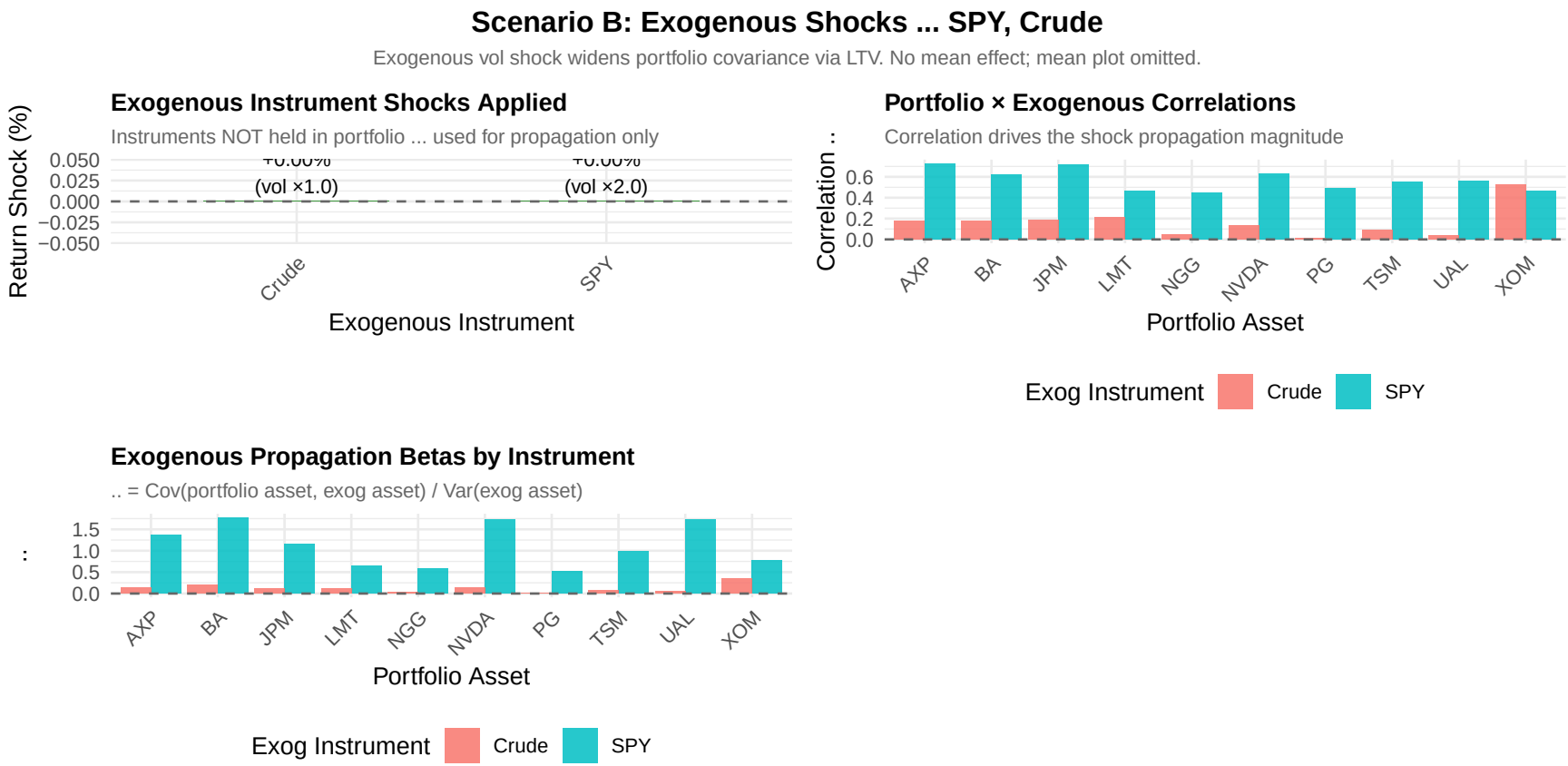

Figure 5: Scenario 3: SPY propagation betas per portfolio asset.

5.3 Portfolio Distribution: Pre vs. Post Stress

```
result_3$stressed_plots$portfolio_distribution_comparison
```

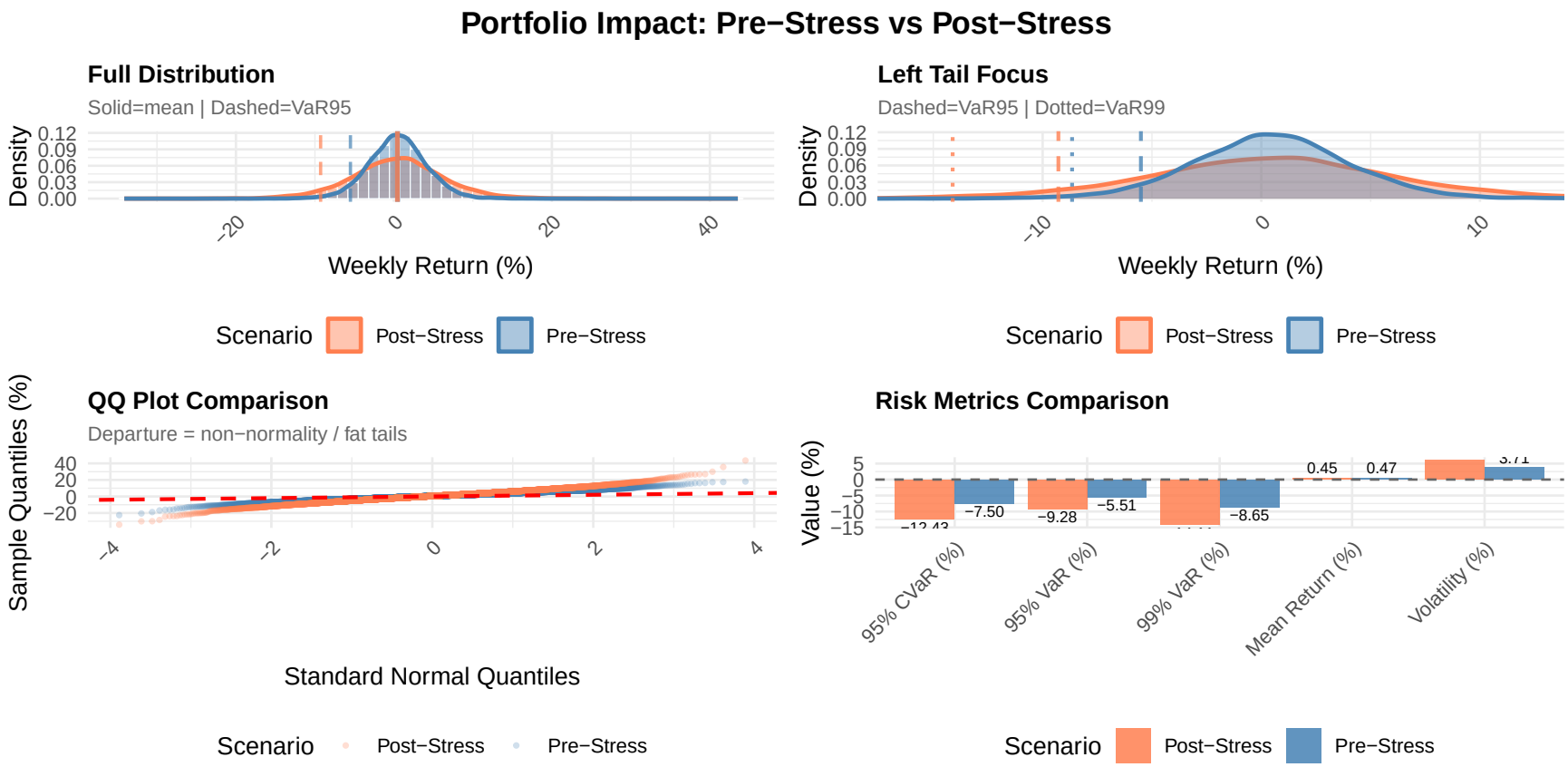

Figure 6: Scenario 3: Portfolio return distribution, base vs. SPY vol x2.

5.4 Stress Impact Summary

Table 4: Scenario 3: Risk metric changes from SPY volatility doubling.

Metric	Pre_Stress	Post_Stress	Change
Expected Return (%)	0.4684	0.4454	-0.0230
Volatility (%)	3.7094	5.9755	2.2662
Skewness	-0.0294	0.0513	0.0807
Excess Kurtosis	1.1019	1.4197	0.3178
VaR 95% (%)	-5.5063	-9.2789	-3.7726
VaR 99% (%)	-8.6528	-14.1143	-5.4615
CVaR 95% (%)	-7.5005	-12.4326	-4.9321

6 Scenario 4: Oil Spike and Market Selloff (Exogenous Return and Vol Shocks)

Shock: SPY falls 5% ($\delta_{SPY} = -0.05$) with vol doubled; Crude surges 20% ($\delta_{Crude} = +0.20$) with vol doubled. This may simulate a stagflationary stress scenario: energy costs surge while equity markets sell off simultaneously.

Both return shocks enter the Schur complement together. The shocked block is $s = \{SPY, Crude\}$ and all portfolio asset means are updated by $\mu_{p|s} = \mu_p + \Sigma_{pe}\Sigma_{ee}^{-1}\delta_e$. Energy-sensitive holdings (XOM) received a net positive effect, whereby a positive effect from a high beta to Crude Oil is partially offset by a negative effect from SPY declines. Broad-market holdings (JPM, AXP, NVDA) are dominated by the SPY decline.

```
exogenous_shock            <- list(SPY = -0.05, Crude = 0.20)
exogenous_volatility_shock <- list(SPY = 2,     Crude = 2)
```

```
result_4 <- portfolio_risk_simulation(
  historical_returns          = returns_data,
  sim_returns_dist            = "rmvt",
  cov_estimation              = "garch",
  calibrate_params            = TRUE,
  calibration_method          = "full",
  copula                      = TRUE,
  copula_type                 = "t",
  copula_df                   = 4,
  n_sim_returns               = NULL,
  propagation                 = TRUE,
  exogenous_returns           = exo_data,
  exogenous_shock             = exogenous_shock,
  exogenous_volatility_shock  = exogenous_volatility_shock,
  df                          = 4,
  seed                        = 123
)
```

6.1 Propagation of the Compound Exogenous Shock

```
result_4$stressed_plots$exogenous_propagation
```

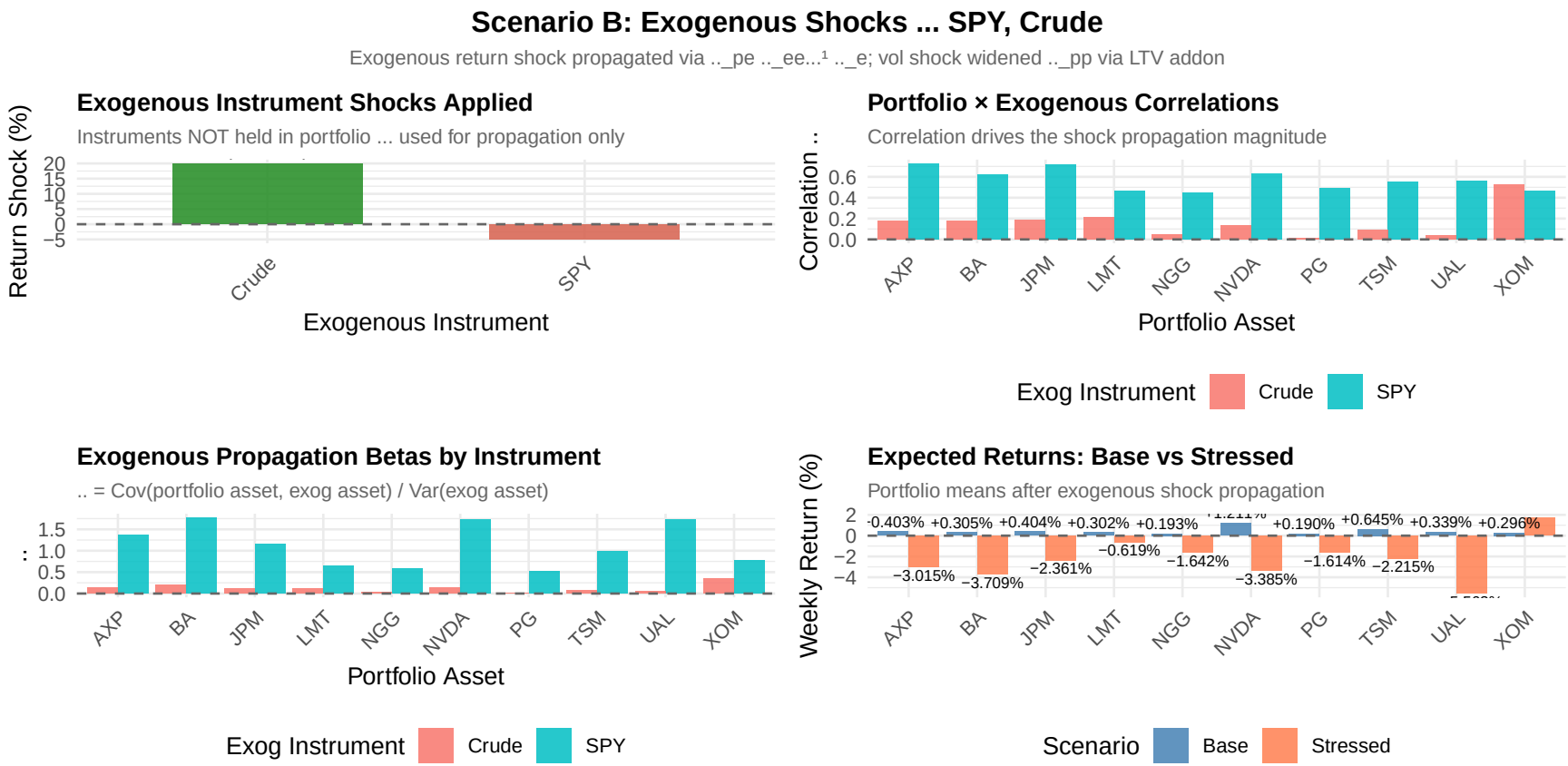

Figure 7: Scenario 4: Conditional mean shifts from SPY -5pct and Crude +20pct.

6.2 Portfolio Distribution Comparison

```
result_4$stressed_plots$portfolio_distribution_comparison
```

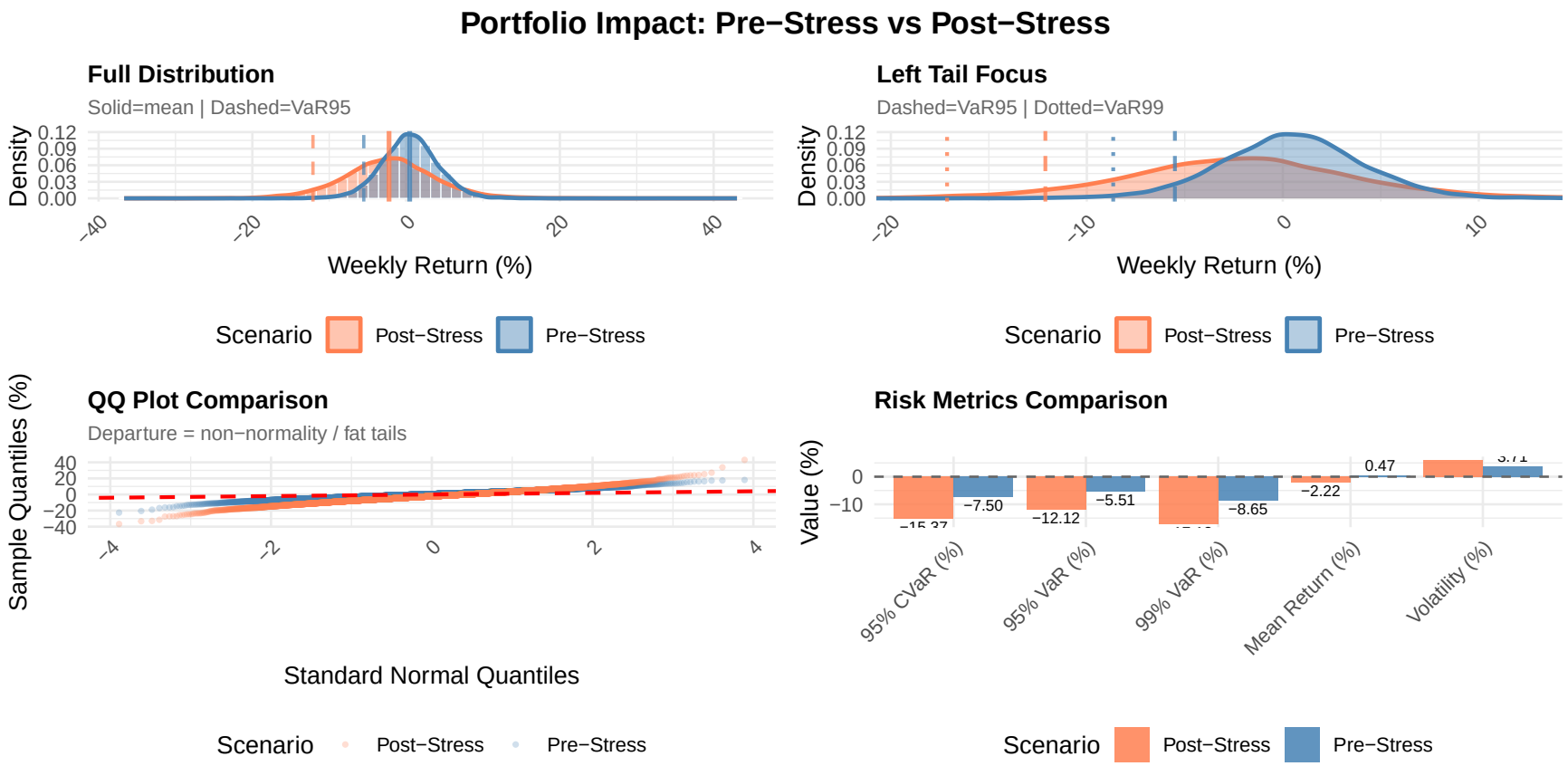

Figure 8: Scenario 4: Portfolio distribution under oil spike and market selloff.

6.3 Stress Impact Summary

Table 5: Scenario 4: Risk metrics under stagflationary shock.

Metric	Pre_Stress	Post_Stress	Change
Expected Return (%)	0.4684	-2.2174	-2.6857
Volatility (%)	3.7094	6.1009	2.3915
Skewness	-0.0294	0.0526	0.0820
Excess Kurtosis	1.1019	1.4224	0.3205
VaR 95% (%)	-5.5063	-12.1157	-6.6094
VaR 99% (%)	-8.6528	-17.1303	-8.4775
CVaR 95% (%)	-7.5005	-15.3732	-7.8727

6.4 Exogenous Historical Correlations

This table explains the propagation magnitudes above. Assets with high SPY correlation (JPM, AXP, NVDA) absorb the largest mean shift from the SPY leg; XOM's positive Crude correlation provides a partial offset.

Table 6: Scenario 4: Historical correlations, portfolio vs. SPY and Crude.

Asset	Crude	SPY
AXP	0.180	0.728
BA	0.182	0.627
JPM	0.188	0.717
LMT	0.217	0.469
NGG	0.049	0.451
NVDA	0.133	0.633
PG	0.017	0.488
TSM	0.092	0.552
UAL	0.036	0.565
XOM	0.526	0.467

7 Scenario 5: Semiconductor Selloff with SPY Offset (Mixed Endo and Exog)

Shock: NVDA and TSM each fall 5% with vol multiplied 1.5x (endogenous return and vol shocks). Simultaneously, SPY rises 2.5% with vol 1.25x (exogenous return and vol shock). This captures a sector-specific stress ie regulatory action or export controls on semiconductors that occurs against a mild broader market rally.

The Schur complement operates on a combined shocked block $s = \{NVDA, TSM, SPY\}$. Assets correlated with the semiconductor pair receive a negative propagated mean shift; the SPY tailwind partially offsets this for holdings with high market beta.

```
asset_shock                <- list(NVDA = -0.05, TSM = -0.05)
volatility_shock           <- list(NVDA = 1.5,   TSM = 1.5)
exogenous_shock            <- list(SPY = 0.025)
exogenous_volatility_shock <- list(SPY = 1.25)
```

```
result_5 <- portfolio_risk_simulation(
  historical_returns          = returns_data,
  sim_returns_dist            = "rmvt",
  cov_estimation              = "garch",
  calibrate_params            = TRUE,
  calibration_method          = "full",
  copula                      = TRUE,
  copula_type                 = "t",
  copula_df                   = 4,
  n_sim_returns               = NULL,
  propagation                 = TRUE,
  asset_shock                 = asset_shock,
  volatility_shock            = volatility_shock,
  exogenous_returns           = exo_data,
  exogenous_shock             = exogenous_shock,
  exogenous_volatility_shock  = exogenous_volatility_shock,
  df                          = 4,
  seed                        = 123
)
```

7.1 Mean Shift Decomposition (Waterfall)

The stacked bar chart decomposes each asset's total conditional mean shift into three components: (1) **direct endogenous shock** (non-zero only for NVDA and TSM), (2) **propagated from endo** (Schur from the NVDA/TSM block), and (3) **propagated from exog** (Schur from the SPY return shock). The diamond marker shows the net total.

```
result_5$stressed_plots$mixed_shock_effects
```

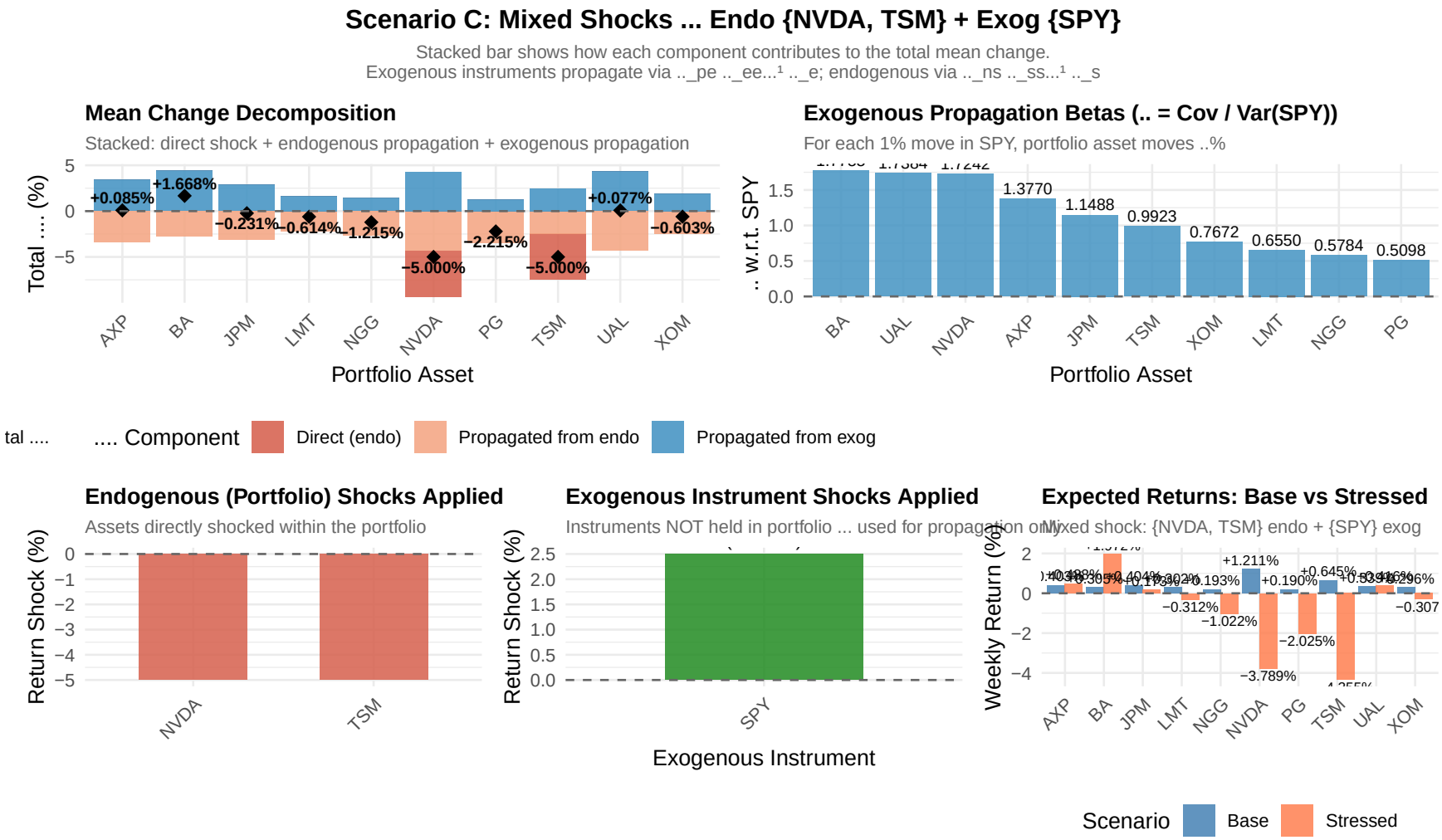

Figure 9: Scenario 5: Mean shift decomposition - direct endo, propagated endo, propagated exog.

7.2 Portfolio Distribution Comparison

Despite the SPY tailwind, the portfolio distribution shifts left because NVDA and TSM are large equal-weight components with high intra-sector correlations that propagate their decline across correlated holdings.

```
result_5$stressed_plots$portfolio_distribution_comparison
```

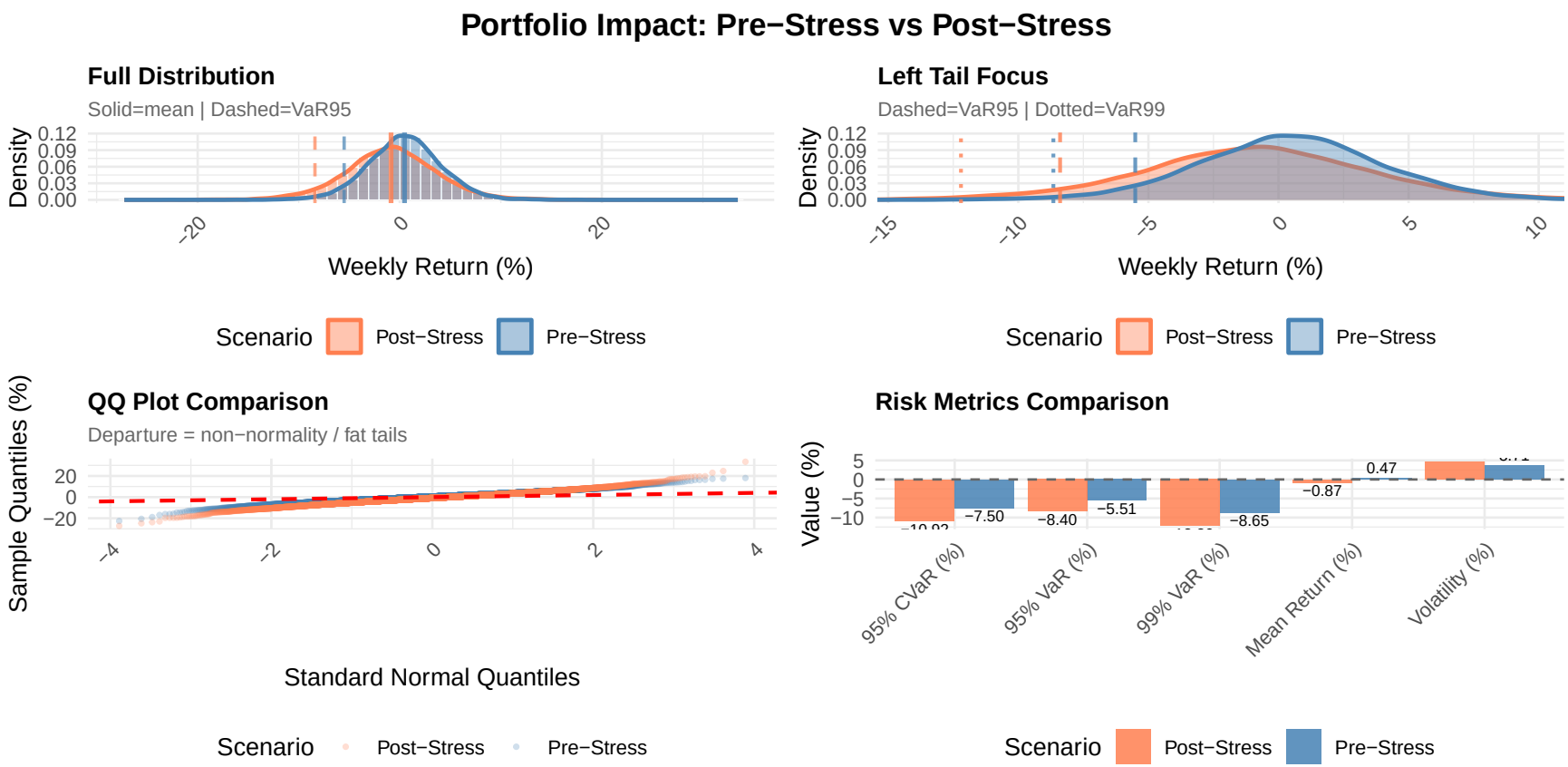

Figure 10: Scenario 5: Portfolio distribution - semiconductor selloff with SPY rally.

7.3 Stress Impact Summary

Table 7: Scenario 5: Risk metrics under semiconductor-specific stress.

Metric	Pre_Stress	Post_Stress	Change
Expected Return (%)	0.4684	-0.8664	-1.3347
Volatility (%)	3.7094	4.6541	0.9447
Skewness	-0.0294	0.0510	0.0804
Excess Kurtosis	1.1019	1.4295	0.3276
VaR 95% (%)	-5.5063	-8.3955	-2.8892
VaR 99% (%)	-8.6528	-12.2048	-3.5520
CVaR 95% (%)	-7.5005	-10.9243	-3.4238

8 Accessing Simulation Draws Directly

The raw per-asset simulated draws are stored as $n_{\text{sim}} \times n_{\text{assets}}$ data frames - accessible without any aggregation to portfolio level. This enables custom downstream analysis such as per-asset VaR.

```
draws_pre  <- result_5$asset_draws
draws_post <- result_5$asset_draws_stress

cat("Pre-stress  draws:", nrow(draws_pre),  "rows x", ncol(draws_pre),  "cols\n")
```

```
## Pre-stress  draws: 10000 rows x 10 cols
```

```
cat("Post-stress draws:", nrow(draws_post), "rows x", ncol(draws_post), "cols\n")
```

```
## Post-stress draws: 10000 rows x 10 cols
```

```
cat("Columns:", paste(names(draws_pre), collapse = ", "), "\n")
```

```
## Columns: AXP, BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM
```

8.1 NVDA: Pre vs. Post Stress Simulated Distribution

```
nvda_comp <- data.frame(
  Return   = c(draws_pre$NVDA, draws_post$NVDA),
  Scenario = rep(c("Pre-Stress","Post-Stress"), each = nrow(draws_pre))
)

means_df <- nvda_comp %>%
  group_by(Scenario) %>%
  summarise(m = mean(Return * 100), .groups = "drop")

ggplot(nvda_comp, aes(x = Return * 100, fill = Scenario)) +
  geom_density(alpha = 0.45, colour = NA) +
  geom_vline(data = means_df, aes(xintercept = m, colour = Scenario),
             linetype = "dashed", linewidth = 0.8) +
  scale_fill_manual(values   = c("Pre-Stress" = "steelblue","Post-Stress" = "coral")) +
  scale_colour_manual(values = c("Pre-Stress" = "steelblue","Post-Stress" = "coral")) +
  labs(title    = "NVDA: Simulated Return Distribution, Pre vs. Post Stress",
       subtitle = "Scenario 5: NVDA direct shock -5%, vol x1.5 | SPY +2.5%, vol x1.25",
       x = "Weekly Return (%)", y = "Density", fill = NULL, colour = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom")
```

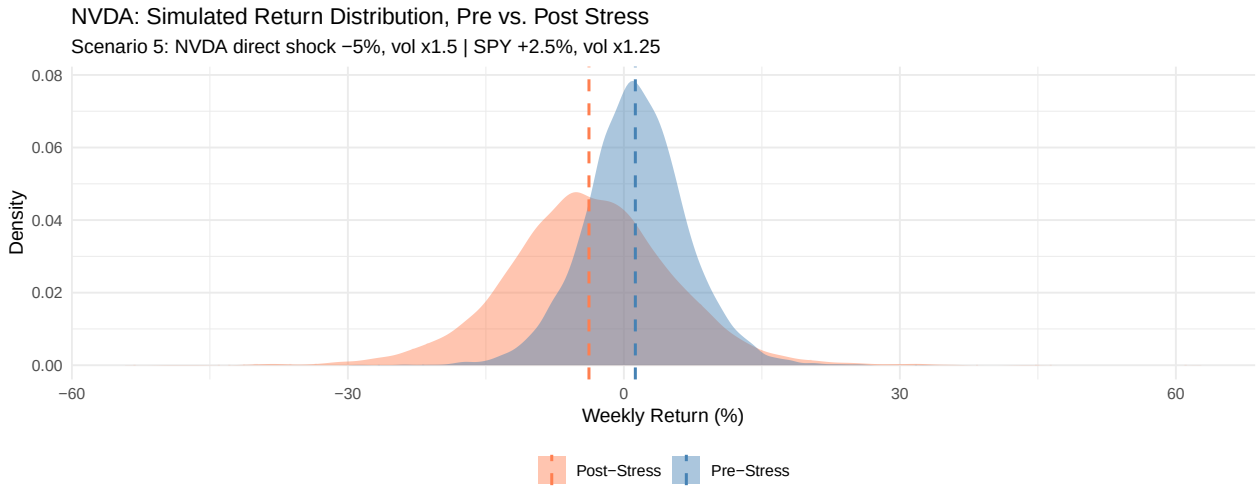

Figure 11: NVDA simulated return distribution, pre and post Scenario 5 stress.

9 Cross-Scenario Comparison

Table 8: Pre-stress portfolio risk statistics across all five scenarios.

Scenario	Mean (%)	Vol (%)	VaR 95% (%)	CVaR 95% (%)
S1: Gaussian, sample cov (no stress)	0.4273	2.9937	-4.4897	-5.7651
S2: t + EWMA + t-copula (no stress)	0.4577	2.4443	-3.4766	-4.8196
S3: t + GARCH, SPY vol x2	0.4684	3.7094	-5.5063	-7.5005
S4: t + GARCH, SPY -5pct / Crude +20pct	0.4684	3.7094	-5.5063	-7.5005
S5: t + GARCH, NVDA/TSM -5pct + SPY +2.5pct	0.4684	3.7094	-5.5063	-7.5005

Scenarios 3–5 show the pre-stress baseline. The incremental stress impact is captured in each scenario's stress impact summary table above.

10 Quick Reference: Result Object Fields

Table 9: Quick reference: result object fields and plot accessors.

Field	Description
result$portfolio_stats	14-row data frame: risk metrics pre and post stress
result$asset_draws	n_sim x n_assets data frame of pre-stress per-asset draws
result$asset_draws_stress	n_sim x n_assets data frame of post-stress per-asset draws
result$asset_empirical_summary	Per-asset empirical mean, SD, skewness, kurtosis, min, max
result$exogenous_historical_correlation	Portfolio x exogenous Pearson correlations (Asset + one col per instrument)
result$exogenous_asset_empirical_summary	Exogenous instrument empirical summary: mean, SD, skewness, kurtosis
result$ks_test	KS test results: Asset, KS_stat, p_value, Flag
result$stress_impact_summary	7-row data frame: Metric, Pre_Stress, Post_Stress, Change
result$stressed_covariance	n x n stressed covariance matrix after shock application
result$stressed_means	Named vector of stressed conditional means per asset
result$final_covariance	n x n final forward-looking covariance matrix
result$copula_correlation	n x n fitted copula correlation matrix
result$tail_dependence	n x n tail dependence matrix (t-copula only)
result$calibrated_params	Calibrated shape parameters per asset
result$plots$...$asset_distributions	Histogram + fitted density overlay, one panel per asset
result$plots$...$portfolio_distribution	Simulated portfolio return distribution (histogram + KDE)
result$plots$...$tail_dependency	Tail dependence bar chart + copula vs. Kendall tau scatter
result$plots$...$time_series	GJR-GARCH conditional vol time series (GARCH cov_estimation only)
result$stressed_plots$portfolio_distribution_comparison	Overlaid pre vs. post stress portfolio return distributions
result$stressed_plots$exogenous_propagation	Exogenous beta and propagation channel plots (Scenario B)
result$stressed_plots$mixed_shock_effects	Waterfall decomposition: direct + propagated endo + exog (Scenario C)

11 Sample Function Output: Console Printout

The `portfolio_risk_simulation` function produces a detailed diagnostic console printout during execution. This section captures a representative example from Scenario 4 (oil spike and market selloff) to illustrate what the output looks like, covering data preprocessing, copula fitting, stress-testing steps, and the final stress impact summary.

historical_returns converted from tibble to data frame Date column detected: 'week' (536 dates, 2016-01-03 to 2026-04-05) Data frequency inferred: Weekly (52 periods/year)

==========
DATA PREPROCESSING SUMMARY ==========
Assets : 10 Asset names : AXP, BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM Observations : 536 (from 536 raw rows) Distribution : rmvt Copula : YES (T) Cov estimation : GARCH

Per-asset empirical summary: AXP: mean=0.0040 sd=0.0442 skew=0.154 kurt=10.207 min=-0.2558 max=0.2883 BA: mean=0.0030 sd=0.0662 skew=1.707 kurt=33.621 min=-0.4418 max=0.7051 JPM: mean=0.0040 sd=0.0374 skew=-0.004 kurt=7.418 min=-0.1964 max=0.2226 LMT: mean=0.0030 sd=0.0327 skew=0.290 kurt=8.251 min=-0.1409 max=0.1963 NGG: mean=0.0019 sd=0.0300 skew=-0.512 kurt=8.667 min=-0.1667 max=0.1575 NVDA: mean=0.0121 sd=0.0637 skew=0.362 kurt=4.498 min=-0.2005 max=0.3019 PG: mean=0.0019 sd=0.0244 skew=-0.178 kurt=5.729 min=-0.1063 max=0.1140 TSM: mean=0.0065 sd=0.0420 skew=0.219 kurt=4.219 min=-0.1393 max=0.1817 UAL: mean=0.0034 sd=0.0719 skew=0.688 kurt=12.252 min=-0.4116 max=0.5125 XOM: mean=0.0030 sd=0.0384 skew=-0.111 kurt=6.673 min=-0.2007 max=0.1674

==========
EXOGENOUS INSTRUMENTS ==========
Count : 2 Names : Crude, SPY Rows matched : 536 (aligned with portfolio observations)

Exogenous empirical summary: Crude: mean=0.0034 sd=0.0574 skew=0.528 kurt=9.998 SPY: mean=0.0024 sd=0.0234 skew=-0.561 kurt=9.422

Portfolio x Exogenous historical correlations: Crude SPY AXP 0.180 0.728 BA 0.182 0.627 JPM 0.188 0.717 LMT 0.217 0.469 NGG 0.049 0.451 NVDA 0.133 0.633 PG 0.017 0.488 TSM 0.092 0.552 UAL 0.036 0.565 XOM 0.526 0.467

==========
CALIBRATING DISTRIBUTION PARAMETERS ==========
Distribution : rmvt Calibration method: FULL

Calibrating degrees of freedom: AXP: df = 8.5000 BA: df = 8.0000 JPM: df = 10.0000 LMT: df = 9.0000 NGG: df = 9.0000 NVDA: df = 10.0000 PG: df = 10.0000 TSM: df = 10.0000 UAL: df = 8.5000 XOM: df = 9.5000

Joint df (mean): 9.2500

==========
CALIBRATION SUMMARY ==========
Asset Optimal_DF Valid AXP 8.5 TRUE BA 8.0 TRUE JPM 10.0 TRUE LMT 9.0 TRUE NGG 9.0 TRUE NVDA 10.0 TRUE PG 10.0 TRUE TSM 10.0 TRUE UAL 8.5 TRUE XOM 9.5 TRUE

Joint df for simulation: 9.2500

rmvt: using calibrated df_joint = 9.2500

Reference asset defaulted to: AXP

==========
PORTFOLIO RISK SIMULATION (WEEKLY RETURNS) ==========
Assets : 10 — AXP, BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM Reference asset : AXP (index 1) Observations : 536 Distribution : rmvt Copula : T Covariance estimation : GARCH Calibrate params : YES Propagation : YES `exogenous_shock` resolved from list (name format): SPY=-0.05, Crude=0.2 `exogenous_volatility_shock` resolved from list (name format): SPY=2, Crude=2

Weights: Equal weighted (1/N) Long-only portfolio

==========
WEIGHTS SUMMARY ==========
Portfolios : 1 Simulations each : 10,000 Total simulations : 10,000 Portfolio type : Long-Only

Portfolio weights: AXP : +0.1000 (+10.00%) BA : +0.1000 (+10.00%) JPM : +0.1000 (+10.00%) LMT : +0.1000 (+10.00%) NGG : +0.1000 (+10.00%) NVDA : +0.1000 (+10.00%) PG : +0.1000 (+10.00%) TSM : +0.1000 (+10.00%) UAL : +0.1000 (+10.00%) XOM : +0.1000 (+10.00%) SUM : +1.0000

==========
FITTING DISTRIBUTIONS ==========

Path: COPULA — marginals + copula separately Copula type: T AXP: t(=0.0040, _scale=0.0386, _marginal=0.0442, df=8.50) BA: t(=0.0030, _scale=0.0573, _marginal=0.0662, df=8.00) JPM: t(=0.0040, _scale=0.0335, _marginal=0.0374, df=10.00) LMT: t(=0.0030, _scale=0.0288, _marginal=0.0327, df=9.00) NGG: t(=0.0019, _scale=0.0264, _marginal=0.0300, df=9.00) NVDA: t(=0.0121, _scale=0.0569, _marginal=0.0637, df=10.00) PG: t(=0.0019, _scale=0.0218, _marginal=0.0244, df=10.00) TSM: t(=0.0065, _scale=0.0376, _marginal=0.0420, df=10.00) UAL: t(=0.0034, _scale=0.0629, _marginal=0.0719, df=8.50) XOM: t(=0.0030, _scale=0.0341, _marginal=0.0384, df=9.50)

==========
MARGINAL FITTING SUMMARY ==========
Distribution : rmvt Assets fitted: 10 Fitted means : 0.0040, 0.0030, 0.0040, 0.0030, 0.0019, 0.0121, 0.0019, 0.0065, 0.0034, 0.0030 Fitted SDs : 0.0386, 0.0573, 0.0335, 0.0288, 0.0264, 0.0569, 0.0218, 0.0376, 0.0629, 0.0341

Estimating GARCH forward volatilities… GARCH estimation complete

Forward-looking volatilities: AXP: 4.45% (historical: 3.86%) BA: 5.78% (historical: 5.73%) JPM: 3.42% (historical: 3.35%) LMT: 3.07% (historical: 2.88%) NGG: 3.71% (historical: 2.64%) NVDA: 5.67% (historical: 5.69%) PG: 3.53% (historical: 2.18%) TSM: 4.25% (historical: 3.76%) UAL: 6.98% (historical: 6.29%) XOM: 3.64% (historical: 3.41%)

Fitting copula for dependence structure… PIT method: fitted RMVT marginals Uniform range: [0.0000, 1.0000] n_obs=536 | n_params=46 | ratio=11.7 → strategy: unstructured + MPL Warm-start : [0.327, 0.949] (from Kendall's) t-copula correlation

not PD — applying nearPD t-copula fitted [unstructured + MPL]. df=8.00 | Log-lik: -2247116418577894635181592796646304459975882094073062457486468460039465759750356734507369482909585 97

Tail Dependence Coefficients: AXP — BA: =0.906, =0.522 AXP — JPM: =0.704, =0.243 AXP — LMT: =0.805, =0.350 AXP — NGG: =0.789, =0.329 AXP — NVDA: =0.789, =0.330 AXP — PG: =0.753, =0.289 AXP — TSM: =0.635, =0.190 AXP — UAL: =0.563, =0.147 AXP — XOM: =0.526, =0.129 BA — JPM: =0.674, =0.218 BA — LMT: =0.727, =0.263 BA — NGG: =0.800, =0.344 BA — NVDA: =0.707, =0.246 BA — PG: =0.465, =0.103 BA — TSM: =0.362, =0.070 BA — UAL: =0.650, =0.200 BA — XOM: =0.570, =0.151 JPM — LMT: =0.626, =0.184 JPM — NGG: =0.676, =0.220 JPM — NVDA: =0.910, =0.531 JPM — PG: =0.509, =0.121 JPM — TSM: =0.756, =0.293 JPM — UAL: =0.787, =0.328 JPM — XOM: =0.775, =0.313 LMT — NGG: =0.584, =0.159 LMT — NVDA: =0.689, =0.230 LMT — PG: =0.791, =0.332 LMT — TSM: =0.580, =0.157 LMT — UAL: =0.693, =0.233 LMT — XOM: =0.775, =0.314 NGG — NVDA: =0.848, =0.412 NGG — PG: =0.535, =0.133 NGG — TSM: =0.368, =0.072 NGG — UAL: =0.750, =0.287 NGG — XOM: =0.484, =0.111 NVDA — PG: =0.709, =0.247 NVDA — TSM: =0.760, =0.297 NVDA — UAL: =0.792, =0.333 NVDA — XOM: =0.705, =0.244 PG — TSM: =0.799, =0.343 PG — UAL: =0.425, =0.089 PG — XOM: =0.443, =0.095 TSM — UAL: =0.388, =0.078 TSM — XOM: =0.515, =0.124 UAL — XOM: =0.747, =0.284

Fitted copula df (8.00) differs from input copula_df (4).

Copula fit: Log-lik=-2247116418577894635181592796646304459975882094073062457486468460039465759750356 73
AIC=4494232837155789270363185593292608919951764188146124914972936920078931519500713469014738965819
BIC=4494232837155789270363185593292608919951764188146124914972936920078931519500713469014738965819

Final Σ not PD — applying nearPD

Using fitted marginal means (garch path)

rmvt scale matrix (df=9.25)

==========
STEP 6 DIAGNOSTICS ==========

Forward vs historical volatilities: AXP: 4.4513% ↑ (historical: 4.4166%) BA: 5.7776% ↓ (historical: 6.6213%) JPM: 3.4173% ↓ (historical: 3.7427%) LMT: 3.0746% ↓ (historical: 3.2658%) NGG: 3.7052% ↑ (historical: 2.9962%) NVDA: 5.6709% ↓ (historical: 6.3667%) PG: 3.5342% ↑ (historical: 2.4390%) TSM: 4.2511% ↑ (historical: 4.1997%) UAL: 6.9793% ↓ (historical: 7.1937%) XOM: 3.6424% ↓ (historical: 3.8367%)

Historical correlation: AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 1.0000 0.6264 0.7614 0.3849 0.3451 0.3552 0.2874 0.3236 0.6676 0.4864 BA 0.6264 1.0000 0.5650 0.4174 0.3757 0.3388 0.2633 0.3384 0.6410 0.4269 JPM 0.7614 0.5650 1.0000 0.3296 0.2495 0.3552 0.2854 0.3334 0.6518 0.4864 LMT 0.3849 0.4174 0.3296 1.0000 0.3187 0.2150 0.3292 0.1552 0.2500 0.4204 NGG 0.3451 0.3757 0.2495 0.3187 1.0000 0.1669 0.4348 0.1875 0.2503 0.2517 NVDA 0.3552 0.3388 0.3552 0.2150 0.1669 1.0000 0.1608 0.5700 0.2676 0.1281 PG 0.2874 0.2633 0.2854 0.3292 0.4348 0.1608 1.0000 0.1919 0.2490 0.1798 TSM 0.3236 0.3384 0.3334 0.1552 0.1875 0.5700 0.1919 1.0000 0.2863 0.1781 UAL 0.6676 0.6410 0.6518 0.2500 0.2503 0.2676 0.2490 0.2863 1.0000 0.3526 XOM 0.4864 0.4269 0.4864 0.4204 0.2517 0.1281 0.1798 0.1781 0.3526 1.0000

Copula correlation: AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 1.0000 0.9059 0.7038 0.8051 0.7887 0.7894 0.7531 0.6350 0.5629 0.5262 BA 0.9059 1.0000 0.6743 0.7266 0.8003

0.7073 0.4653 0.3616 0.6498 0.5699 JPM 0.7038 0.6743 1.0000 0.6256 0.6757 0.9098 0.5087 0.7562 0.7875 0.7751 LMT 0.8051 0.7266 0.6256 1.0000 0.5840 0.6891 0.7909 0.5804 0.6930 0.7753 NGG 0.7887 0.8003 0.6757 0.5840 1.0000 0.8478 0.5348 0.3678 0.7505 0.4844 NVDA 0.7894 0.7073 0.9098 0.6891 0.8478 1.0000 0.7094 0.7599 0.7919 0.7054 PG 0.7531 0.4653 0.5087 0.7909 0.5348 0.7094 1.0000 0.7993 0.4246 0.4433 TSM 0.6350 0.3616 0.7562 0.5804 0.3678 0.7599 0.7993 1.0000 0.3885 0.5147 UAL 0.5629 0.6498 0.7875 0.6930 0.7505 0.7919 0.4246 0.3885 1.0000 0.7475 XOM 0.5262 0.5699 0.7751 0.7753 0.4844 0.7054 0.4433 0.5147 0.7475 1.0000

Difference (Copula − Historical): AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 0.0000 0.2795 -0.0575 0.4201 0.4436 0.4342 0.4657 0.3115 -0.1046 0.0397 BA 0.2795 0.0000 0.1093 0.3092 0.4246 0.3685 0.2020 0.0232 0.0087 0.1430 JPM -0.0575 0.1093 0.0000 0.2960 0.4263 0.5546 0.2233 0.4228 0.1357 0.2887 LMT 0.4201 0.3092 0.2960 0.0000 0.2653 0.4741 0.4618 0.4252 0.4430 0.3549 NGG 0.4436 0.4246 0.4263 0.2653 0.0000 0.6809 0.1000 0.1804 0.5001 0.2327 NVDA 0.4342 0.3685 0.5546 0.4741 0.6809 0.0000 0.5486 0.1900 0.5243 0.5773 PG 0.4657 0.2020 0.2233 0.4618 0.1000 0.5486 0.0000 0.6074 0.1756 0.2635 TSM 0.3115 0.0232 0.4228 0.4252 0.1804 0.1900 0.6074 0.0000 0.1022 0.3366 UAL -0.1046 0.0087 0.1357 0.4430 0.5001 0.5243 0.1756 0.1022 0.0000 0.3949 XOM 0.0397 0.1430 0.2887 0.3549 0.2327 0.5773 0.2635 0.3366 0.3949 0.0000 Max |diff|: 0.6809

Final Σ: AXP BA JPM LMT NGG NVDA PG TSM UAL XOM AXP 0.001981 0.002330 0.001071 0.001102 0.001301 0.001993 0.001185 0.001202 0.001749 0.000853 BA 0.002330 0.003338 0.001331 0.001291 0.001713 0.002317 0.000950 0.000888 0.002620 0.001199 JPM 0.001071 0.001331 0.001168 0.000657 0.000856 0.001763 0.000614 0.001099 0.001878 0.000965 LMT 0.001102 0.001291 0.000657 0.000945 0.000665 0.001202 0.000859 0.000759 0.001487 0.000868 NGG 0.001301 0.001713 0.000856 0.000665 0.001373 0.001781 0.000700 0.000579 0.001941 0.000654 NVDA 0.001993 0.002317 0.001763 0.001202 0.001781 0.003216 0.001422 0.001832 0.003134 0.001457 PG 0.001185 0.000950 0.000614 0.000859 0.000700 0.001422 0.001249 0.001201 0.001047 0.000571 TSM 0.001202 0.000888 0.001099 0.000759 0.000579 0.001832 0.001201 0.001807 0.001153 0.000797 UAL 0.001749 0.002620 0.001878 0.001487 0.001941 0.003134 0.001047 0.001153 0.004871 0.001900 XOM 0.000853 0.001199 0.000965 0.000868 0.000654 0.001457 0.000571 0.000797 0.001900 0.001327

Condition number : 149180427.48 Min eigenvalue : 1.53e-10

Uniformity diagnostic (KS test): Asset KS_stat p_value Flag AXP 0.0673 0.0157 REJECT BA 0.0988 0.0001 REJECT JPM 0.0424 0.2896 OK LMT 0.0589 0.0486 REJECT NGG 0.0519 0.1113 OK NVDA 0.0339 0.5694 OK PG 0.0489 0.1540 OK TSM 0.0364 0.4779 OK UAL 0.0708 0.0093 REJECT XOM 0.0633 0.0271 REJECT

Tail dependence retrieved from Step 5 (df=8.0000)

Pair _copula
AXP — BA 0.9059 0.5218 HIGH
AXP — JPM 0.7038 0.2426
AXP — LMT 0.8051 0.3500 HIGH
AXP — NGG 0.7887 0.3294 HIGH
AXP — NVDA 0.7894 0.3303 HIGH
AXP — PG 0.7531 0.2893
AXP — TSM 0.6350 0.1900
AXP — UAL 0.5629 0.1471
AXP — XOM 0.5262 0.1289
BA — JPM 0.6743 0.2184

Pair _copula
BA — LMT 0.7266 0.2631
BA — NGG 0.8003 0.3438 HIGH
BA — NVDA 0.7073 0.2455
BA — PG 0.4653 0.1034
BA — TSM 0.3616 0.0701
BA — UAL 0.6498 0.2002
BA — XOM 0.5699 0.1508
JPM — LMT 0.6256 0.1838
JPM — NGG 0.6757 0.2195
JPM — NVDA 0.9098 0.5307 HIGH
JPM — PG 0.5087 0.1211
JPM — TSM 0.7562 0.2926
JPM — UAL 0.7875 0.3279 HIGH
JPM — XOM 0.7751 0.3134 HIGH
LMT — NGG 0.5840 0.1586
LMT — NVDA 0.6891 0.2302
LMT — PG 0.7909 0.3321 HIGH
LMT — TSM 0.5804 0.1565
LMT — UAL 0.6930 0.2334
LMT — XOM 0.7753 0.3136 HIGH
NGG — NVDA 0.8478 0.4116 HIGH
NGG — PG 0.5348 0.1330
NGG — TSM 0.3678 0.0718
NGG — UAL 0.7505 0.2866
NGG — XOM 0.4844 0.1108
NVDA — PG 0.7094 0.2474
NVDA — TSM 0.7599 0.2966
NVDA — UAL 0.7919 0.3333 HIGH
NVDA — XOM 0.7054 0.2440
PG — TSM 0.7993 0.3425 HIGH
PG — UAL 0.4246 0.0889
PG — XOM 0.4433 0.0953
TSM — UAL 0.3885 0.0777
TSM — XOM 0.5147 0.1237
UAL — XOM 0.7475 0.2835

Max |latent_cor − copula_cor|: 0.6883

base_sigma not PD before stress testing — applying nearPD

Final Σ condition number : 149180429.30 Final Σ min eigenvalue : 1.53e-10

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STRESS TESTING ==========
Portfolio return-shocked : 0 — none Portfolio vol-shocked : 0 — none Exogenous return-shocked :
2 — SPY, Crude Exogenous vol-shocked : 2 — SPY, Crude Propagation : ENABLED

— EXOGENOUS VOLATILITY SHOCK — Exogenous Asset Multiplier Crude 2.0000x SPY 2.0000x Exogenous covariance blocks scaled

— RETURN SHOCK & PROPAGATION —

Exogenous return shocks (zero portfolio weight — not simulated): SPY: -0.0500 (-5.00%) Crude: +0.2000 (+20.00%)

Conditional mean propagation: Asset Base Stressed Δ Type ———————————————————————— AXP 0.00403 -0.03015 -0.03418 [propagated] BA 0.00305 -0.03709 -0.04014 [propagated] JPM 0.00404 -0.02361 -0.02765 [propagated] LMT 0.00302 -0.00619 -0.00921 [propagated] NGG 0.00193 -0.01642 -0.01835 [propagated] NVDA 0.01211 -0.03385 -0.04596 [propagated] PG 0.00190 -0.01614 -0.01804 [propagated] TSM 0.00645 -0.02215 -0.02861 [propagated] UAL 0.00339 -0.05563 -0.05902 [propagated] XOM 0.00296 0.01741 0.01445 [propagated] Exogenous (not in portfolio, shown for reference): SPY 0.00244 -0.04756 -0.05000 [EXOG SHOCKED] Crude 0.00335 0.20335 0.20000 [EXOG SHOCKED]

— EXOGENOUS VOLATILITY ADDON (LTV) — Crude: k_e=2.00 → (k_e^2−1)=3.0000 × outer product added SPY: k_e=2.00 → (k_e^2−1)=3.0000 × outer product added Asset marginal vols: pre=4.451%, 5.778%, 3.417%, 3.075%, 3.705%, 5.671%, 3.534%, 4.251%, 6.979%, 3.642% | post=7.264%, 9.455%, 5.897%, 4.241%, 4.390%, 9.110%, 4.093%, 5.886%, 9.920%, 5.925% sigma_stressed not PD — applying nearPD

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STRESSED PARAMETERS SUMMARY ==========

Asset _base _stressed Δ
AXP 0.00403 -0.03015 -0.03418 [propagated]
BA 0.00305 -0.03709 -0.04014 [propagated]
JPM 0.00404 -0.02361 -0.02765 [propagated]
LMT 0.00302 -0.00619 -0.00921 [propagated]
NGG 0.00193 -0.01642 -0.01835 [propagated]
NVDA 0.01211 -0.03385 -0.04596 [propagated]
PG 0.00190 -0.01614 -0.01804 [propagated]
TSM 0.00645 -0.02215 -0.02861 [propagated]
UAL 0.00339 -0.05563 -0.05902 [propagated]
XOM 0.00296 0.01741 0.01445 [propagated]

Asset _base _stressed Δ
AXP 4.4513 7.2635 2.8122 [propagated]
BA 5.7776 9.4548 3.6772 [propagated]
JPM 3.4173 5.8971 2.4798 [propagated]
LMT 3.0746 4.2411 1.1666 [propagated]
NGG 3.7052 4.3900 0.6848 [propagated]
NVDA 5.6709 9.1104 3.4394 [propagated]
PG 3.5342 4.0930 0.5588 [propagated]
TSM 4.2511 5.8861 1.6350 [propagated]
UAL 6.9793 9.9201 2.9408 [propagated]
XOM 3.6424 5.9249 2.2825 [propagated]

Asset	_base	_stressed	Δ

Exogenous shock applied (zero weight — not simulated): SPY Δret=-0.0500 vol_mult=2.0000x Crude Δret=+0.2000 vol_mult=2.0000x

sigma_stressed condition number : 149884300.87 sigma_stressed min eigenvalue : 4.10e-10

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SIMULATING PORTFOLIO RETURNS ==========
base_sigma not PD — applying nearPD sigma_stressed not PD — applying nearPD rmvt: deflating covariance to scale matrix (df=9.2500) rmvt df: 9.2500

Simulating 10,000 scenarios across 1 weight set(s)…

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SIMULATION SUMMARY ==========
Distribution : rmvt Simulation df : 9.2500 Weight sets : 1 Sims per set : 10,000 Valid pre-obs : 10,000 / 10,000 Valid post-obs : 10,000 / 10,000

Sanity check: Portfolio Target: 0.4287% Simulated: 0.4684% Portfolio Target: 3.7303% Simulated: 3.7094% Simulated vol consistent with target

==========
PRE-STRESS PORTFOLIO STATISTICS (Weekly Returns) ==========

Sample size: 10,000

RETURN (Weekly) Expected Return (mean) : +0.4684% Expected Return (median) : +0.4830%

RETURN (Annualised) Annual Return (geometric) : +27.5050% Annual Volatility : 26.7485% Sharpe Ratio (annual) : 0.9105

RISK (Weekly) Volatility : 3.7094% Skewness : -0.0294 Excess Kurtosis : 1.1019

VALUE-AT-RISK (Weekly, negative = loss) 90% VaR (10th pct) : -4.0538% 95% VaR (5th pct) : -5.5063% 99% VaR (1st pct) : -8.6528%

EXPECTED SHORTFALL (Weekly) 90% CVaR : -6.1060% 95% CVaR : -7.5005% 99% CVaR : -10.9240%

TAIL DIAGNOSTICS % Positive months : 55.34% % Negative months : 44.66% Worst single return : -22.5812% Best single return : +18.0529%

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POST-STRESS PORTFOLIO STATISTICS (Weekly Returns) ==========

Sample size: 10,000

RETURN (Weekly) Expected Return (mean) : -2.2174% Expected Return (median) : -2.1856%

RETURN (Annualised) Annual Return (geometric) : -68.8390% Annual Volatility : 43.9941% Sharpe Ratio (annual) : -2.6209

RISK (Weekly) Volatility : 6.1009% Skewness : 0.0526 Excess Kurtosis : 1.4224

VALUE-AT-RISK (Weekly, negative = loss) 90% VaR (10th pct) : -9.6567% 95% VaR (5th pct) : -12.1157% 99% VaR (1st pct) : -17.1303%

EXPECTED SHORTFALL (Weekly) 90% CVaR : -13.0894% 95% CVaR : -15.3732% 99% CVaR : -20.4826%

TAIL DIAGNOSTICS % Positive months : 34.16% % Negative months : 65.84% Worst single return : -36.6715% Best single return : +43.0367%

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STRESS IMPACT SUMMARY ==========

Metric	Pre-Stress	Post-Stress	Change
Expected Return	0.4684%	-2.2174%	-2.6857%
Volatility	3.7094%	6.1009%	+2.3915%
Skewness	-0.0294	0.0526	+0.0820
Excess Kurtosis	1.1019	1.4224	+0.3205
95% VaR	-5.5063%	-12.1157%	-6.6094%
99% VaR	-8.6528%	-17.1303%	-8.4775%
95% CVaR	-7.5005%	-15.3732%	-7.8727%

Warning: VaR95 worsened by 6.61% Warning: CVaR95 worsened by 7.87% Warning: Vol increased by 2.39%

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GENERATING PORTFOLIO DIAGNOSTIC PLOTS ==========
Plot 1: Asset distributions Plot 2: Portfolio distribution Plot 3: QQ plot Plot 4: Historical correlation Plot 5: Copula correlation + difference Plot 6: Covariance matrix

Building GARCH conditional volatility time-series plots… Plot 7: GARCH time-series (10 asset(s), 4 col × 3 row canvas)

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GENERATING PAIRWISE COPULA DIAGNOSTIC PLOTS Reference asset: AXP ==========
Pairs: AXP vs BA, JPM, LMT, NGG, NVDA, PG, TSM, UAL, XOM Plot 11: Pairwise diagnostics (9 pairs, 4-column layout)

Building tail dependency & correlation divergence plots… Plot 11b: Tail dependency & correlation divergence

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CREATING STRESS PROPAGATION PLOTS ==========
Scenario : B: Exogenous (non-portfolio instruments) Exog shocked (2) : SPY, Crude Plot 12b: Scenario B — exogenous-only propagation

Building pre/post stress distribution comparison… Plot 12c: Pre/post distribution comparison

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COMPILING RESULTS ==========

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SIMULATION COMPLETE ==========

Pre-stress samples : 10,000 Post-stress samples : 10,000 Diagnostic plots : 10 Stress plots : 2